Chapter 13 – Building Information Systems

Fall 2025 – Professor Ralph Marra

Chapter 13 Learning Objectives

- How does building new systems produce organizational change?
- What are the core activities in the systems development process?
- What are the principal methodologies for modeling and designing systems?
- What are alternative methods for building information systems?
- What are new approaches for system building in the digital firm era?

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Students – please read this Case Study – next 3 slides - (take 5 minutes)

Vinci Energies Builds a New Mobile Employee Time Sheet App

Vinci Energies, part of the Vinci Group and headquartered in France, is a growing professional services company for the construction, energy, and information technology industries. It provides solutions for infrastructure, building, and information and communications technology (ICT), and develops and maintains efficient energy infrastructures. The company had 1,800 business units with over 82,000 employees in 55 countries and global revenue of 13.8 billion Euros (U.S. $15 billion) in 2019.

Vinci had pursued a strong acquisition strategy, tripling in size between 2009 and 2019. Vinci moved all of its companies worldwide to a single instance of SAP S/4HANA, SAP's next generation of ERP software, so that the entire enterprise would operate on a common set of business processes. The company was able to migrate to the new SAP S/4HANA system in August 2018, and wanted additional applications created using the SAP platform.

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What Vinci produces are actually services, so the business is entirely project-based, and each project has its own financial arrangements with clients. Every cost for the company (such as hours worked or equipment rental) is tied to a specific project, and a single employee might be working on multiple projects. If an employee in the field works on more than one project on a given day, that person might need to input 2 hours spent on one project and 4 hours 15 minutes spent on another. That person might also have to add equipment rental costs for one of these projects. Management uses this information to determine whether a project is earning or losing money for the business. Vinci Energies manages about 400,000 projects worldwide each year, each valued at about 20,000 to 25,000 Euros (U.S. $22,000 to $27,000).

Vinci's existing systems, which were heavily manual, made it very difficult and time-consuming to keep track of all these project costs. Employees working in the field (many of whom traveled to perform construction and maintenance at remote sites) would have to return to the office to enter project and time sheet data online into the SAP system. The company needed a new employee time sheet application that could capture worker time sheets as well as project costs and activities for multiple projects from any location. The time sheet and project data would be able to be viewed by other authorized users online.

Vinci undertook a pilot project using SAP Cloud Platform for this purpose. SAP Cloud Platform is an enterprise Platform-as-a-Service (PaaS), which lets organizations build, extend, and integrate business applications in the cloud. It has an intuitive user interface and simple development tools. Vinci developed an app called MOBITIME, using this tool within a seven-month period. Business and application users spent considerable time discussing business processes and data entry screen design to make sure the system had the correct functionality, the design was acceptable, and people were comfortable using the software.

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MOBITIME is only available online, but employees can easily access it on tablets and mobile phones when they are in the field. They are able to enter time sheet data or other project costs using a single application, which transmits the data to update the new SAP S/4HANA system. MOBITIME's interface is easily navigable and allows employees to input all the data in one place. After transmission to the SAP S/4HANA system, the data can be viewed at the back office to help project managers keep track of their projects' various costs.

Sources: www.vinci.com, accessed April 8, 2020; Lauren Bonneau, VINCI Energies Simplifies Employees' Lives with New Timesheet-Entry Application," SAPInsider, February 28, 2019; www.sap.com, accessed April 9, 2020; and Lauren Meyerhoff, "With an Aggressive Acquisition Strategy, Vinci Energies Triples Its Size In 10 Years," *Forbes*, September 5, 2019.

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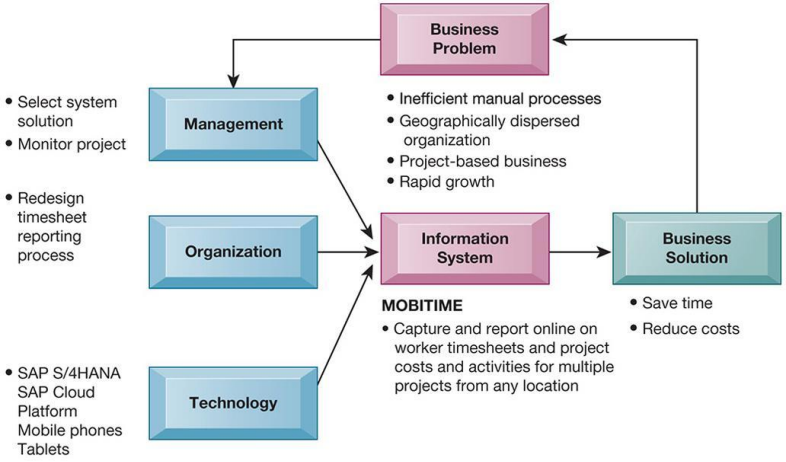
Vinci Energies Builds a New Mobile Employee Time Sheet App

- ***Problem***
  - Inefficient manual processes
  - Geographically dispersed organization
  - Project-based business
  - Rapid growth
- ***Solutions***
  - Select system solution
  - Monitor project
  - Redesign timesheet reporting process
  - SAP S/4 Hana; SAP Cloud Platform
  - Mobile phones, tablets
  - New MOBTIME app
  - Demonstrates steps needed to design and build new information systems
  - Illustrates redesigning business processes as a part of system building



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Systems Development and Organizational Change

- **IT-enabled Organizational Change**

- **Automation (When Is Automation Better Than Manual Processing?)**
  - ✓ Increases efficiency
  - ✓ Replaces manual tasks

- **Rationalization of Procedures (Are Procedures Helping or Hurting Organizations?)**
  - ✓ Streamlines standard operating procedures
  - ✓ Often found in programs for making continuous quality improvements
    - Total quality management (TQM)
    - Six Sigma

- **Business Process Redesign (Are Procedures Optimized? How do you know?)**
  - ✓ Analyze, simplify, and redesign business processes
  - ✓ Reorganize workflow, combine steps, eliminate repetition

- **Paradigm Shifts (Technology is forcing changes in the business landscape)**
  - ✓ Rethink nature of business
  - ✓ Define new business model
  - ✓ Change nature of organization

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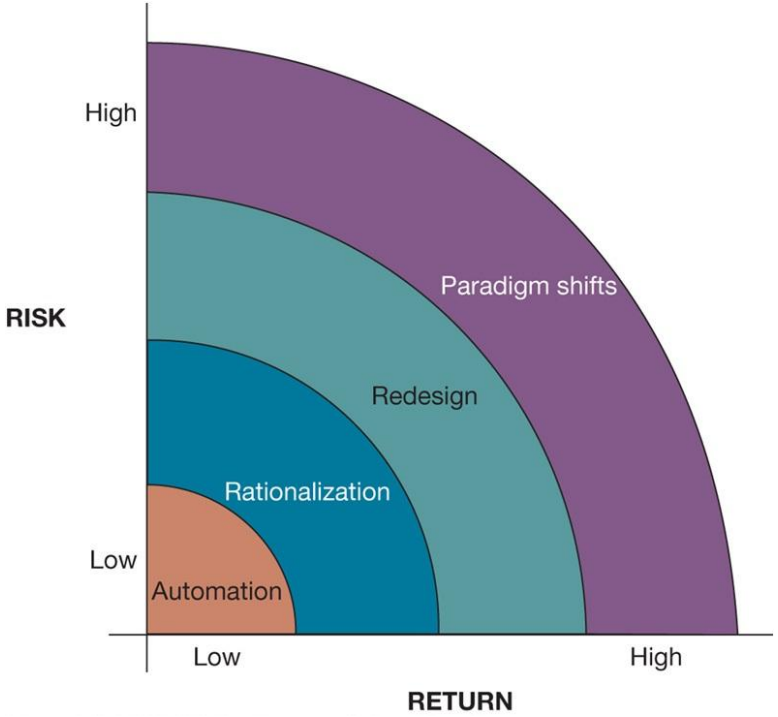
Organizational Change Carries Risks and Rewards



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Interactive Session Organizations

Tommy Hilfiger Transforms Its Wholesale Sales Process with Digital Showrooms

Tommy Hilfiger's world-leading designer apparel and retail brand is known for its "preppy with a twist" designs. The company maintains the Tommy Hilfiger and Hilfiger Denim brands, which include Hilfiger Collection, Tommy Hilfiger Tailored, men's, women's, and children's sportswear, denim, accessories, and footwear. Tommy Hilfiger was acquired by PVH Corp. in 2010 and has an extensive distribution network in more than 100 countries and 2,000 retail stores throughout North America, Europe, Latin America, and the Asia Pacific region. Global retail sales of the Tommy Hilfiger brand were $9.2 billion in 2019.

Hilfiger's management wanted the firm to take better advantage of digital technology to improve the way it operated. For example, the wholesale sales process cried out for streamlining and simplification. In the past, several times per year, buyers of Hilfiger merchandise for retail stores and department store chains would have to personally visit a Hilfiger wholesale showroom, which displayed physical samples of each item in the Hilfiger product line for the upcoming sales season. The buyers would inspect physical samples from a new collection and write down orders on paper order forms. The entire process of viewing and ordering items to stock retail stores might take three days.

Tommy Hilfiger encompasses eight sub-brands, each selling 1,500 pieces of clothing every quarter. That translates into a great deal of product to create just to show and sell samples to a wholesale buyer and then produce them to sell to the end customer.

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Hilfiger assembled a small team of information technology specialists and a designated "digital transformation team" to reimagine the wholesale sales process. The team spent several weeks in Hilfiger showrooms observing how sales people worked with retail and department store buyers. The team recommended streamlining the wholesale ordering process by using digital showrooms to shorten the window between retailer previews of new collections and final delivery of new products to stores. The digital showrooms feature interactive touchscreen workstations with ultra-high-definition display screens where buyers can view every single item in the Tommy Hilfiger collection. They can zoom in to see design details and textures, and click every garment to display prices, color options, and size ranges, and place orders. At the end of the process, buyers receive an email with an attached PDF document containing their final order. The system arranges for order delivery.

Tommy Hilfiger used the Couchbase Data Platform as the technology foundation for its digital showroom. Couchbase uses NoSQL nonrelational data management technology (see **Chapter 6**), which provides more flexibility and scalability than traditional data management tools. Compared to traditional tools, Couchbase can more easily handle large numbers of interactions with end users and different kinds of data. It is optimized for interactive applications and rich, personalized customer experiences and engagement, such as wholesale buyers scanning new product lines, selecting products, examining texture and design features, and inquiring about pricing and shipment dates. Workstation users can add, access, and combine data in real time.

The digital showroom system has minimized production and transport of showroom samples because there is much less need to create, examine, and deliver samples to showroom locations around the world for every new collection. Using the digital showroom, Tommy Hilfiger was able to cut sample production by 80 percent in European headquarters, and it's doing the same with its digital showrooms all over the world. Costs to transport physical samples have also been reduced.

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When Hilfiger's Asia-Pacific team went on a buying trip to Europe, the visit only took a single day, as opposed to three days in the past. Pre-Fall sales increases were noted for the Netherlands, Africa, and Middle East. Shifting from showrooms full of physical samples to digital showrooms has reduced the amount of time required to provide new products to a retail outlet for sale to the public by six weeks. The digital showrooms have also been able to accommodate more wholesale buyer appointments per day, and sales are much more accurate.

Hilfiger launched its first digital showroom in Amsterdam in 2015. By mid-2019, PVH had launched 53 digital showrooms in countries around the globe, featuring over 300,000 products.

Tommy Hilfiger's success with digital showrooms has encouraged it to look for other opportunities for business process transformation in its value chain. The company is now transitioning to 3-D design for designing and manufacturing garments. All of Tommy Hilfiger's design processes—from sketching through sampling and showrooming—will be digital.

Sources: global.tommy.com, accessed April 8, 2020; "This Season's Must-Have," www.couchbase.com, accessed February 26, 2020; Maghan McDowell, "Tommy Hilfiger Goes All In on Digital Design," *Vogue Business*, November 7, 2019; Alex Hisaka, "Tommy Hilfiger Revolutionizes the Fashion Industry with a Digital Selling Ecosystem Powered by DC/OS," D2IQ, December 10, 2019; and Chris Preimesberger, "IT Science Case Study: How Tommy Hilfiger Created Its Digital Showroom," *eWeek*, July 11, 2018.

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Tommy Hilfiger Transforms Its Wholesale Sales Process with Digital Showrooms

1. **How did Tommy Hilger's previous wholesale sales process affect its business performance?**

2. **What management, organization, and technology factors contributed to Tommy Hilfiger's problems with its wholesale sales business process?**

3. **Describe the role of technology and digital showrooms in Tommy Hilfiger's business process changes.**

4. **How did Tommy Hilfiger's redesigned business process change the way the company worked? What was their business impact? Explain.**

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Tommy Hilfiger Transforms Its Wholesale Sales Process with Digital Showrooms

1. **How did Tommy Hilger's previous wholesale sales process affect its business performance?**

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2. **What management, organization, and technology factors contributed to Tommy Hilfiger's problems with its wholesale sales business process?**

   - **Management**: Hilfiger's management would have to arrange for in person visits with clients which would cause delays in selling and rolling out new product to these customers and created some logistical challenges with travel.
   - **Organization**: The organization's size and reach made this an inefficient process given that it built in delays throughout their production process.
   - **Technology**: Technology did not necessarily contribute to the problem, rather the lack of the use of available remote technologies made the entire process and effort take longer.

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Tommy Hilfiger Transforms Its Wholesale Sales Process with Digital Showrooms

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Business *Process* Management (BPM)

- Variety of tools, methodologies to analyze, design, optimize ***processes***
- Used by firms to manage business ***process redesign***

- Steps in BPM

1. Identify processes for change
2. Analyze existing processes
3. Design the new process
4. Implement the new process
5. Continuous measurement

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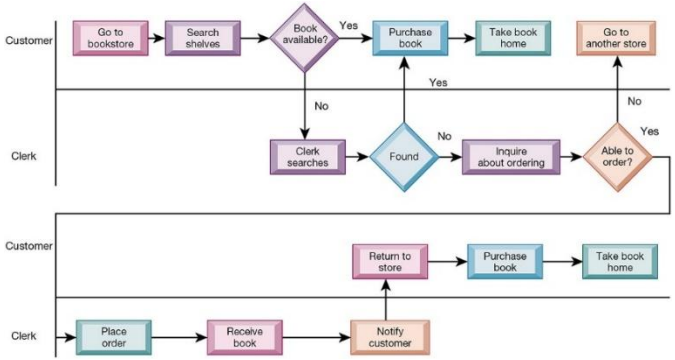
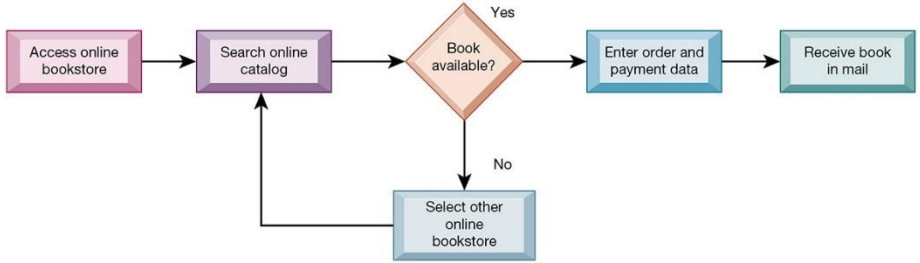
Automate a Process

Business Process for Purchasing a Book from a Physical Bookstore

1 - Review, Analyze and Understand The Current Process:



2 – Design The New/Automated Process:



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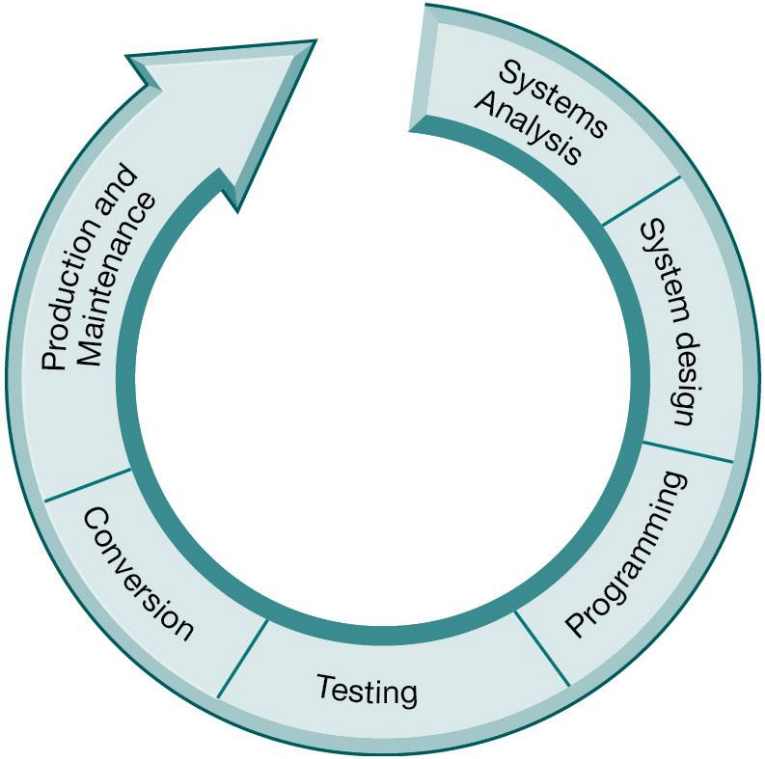
The Systems Development Life Cycle (SDLC)



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6 Key Phases of SDLC

1. **System Analysis** – Understand and document the problem/solution

2. **System Design** – Design an information system (IS) to align with the solution

3. **System Development** – Start building/programming the IS solution

4. **Testing** – Test IS solution in both isolated (unit) and parallel (end to end) modes

5. **Conversion** – Implement and replace manual processes with IS solution

6. **a) Production** – Go Live with new system
   b) Maintenance – Continually optimize and fix issues (bugs) in new implementation

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Phase 1 - Systems Analysis

- Analysis/***Documentation*** of problem to be solved by new system
  - Defining the problem
  - Identifying causes
  - Specifying solutions
  - Identifying information requirements
  - Diagram all workflows
- Feasibility study (manpower, costs**, time**, training) – go/no-go decision
- If feasible - Systems proposal report (software dev required, vendors, costs, special type of programmers, data (databases), reports)
- If proposal approved - Information requirements **(left off here)**

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Specific information requirements must be established – OTHERWISE…

- ✓ If done improperly - you may be building a system that doesn't solve an organization's problem. ***Faulty requirements analysis*** is a **leading cause** of systems failure and high systems development costs.

- ✓ ***Insufficient user involvement*** in the design effort is the next leading cause of erroneous systems development. Too many times system analysis and design are left up to the IT staff instead of involving users throughout the system development process.

- ✓ **BPM Documentation not existent or incomplete**

- ✓ ***People don't like change*** and will sometimes do what they can to prevent it. Any changes introduced into the organization will be the most problematic to manage.

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Phase 2 - Systems Design

- Identify and ***document*** existing processes (**this should be readily available via BPM**)
- ***Document*** system specifications to deliver functions identified during systems analysis (Phase 1) – document any changes to existing processes that need to be made.
- Identify ***inefficiencies*** (what was not working)
- Create ***models of improved processes*** (samples of how each process will change)
- Capture and enforce ***business rules*** for performing, automating processes
- ***Integrate/Remove*** existing systems to support process improvements (what will be eliminated?)
- Address all ***managerial, organizational, and technological components*** of system solution
- Role of End Users
  - User ***information requirements*** drive system building
  - Users must have ***sufficient control over design process*** to ensure system reflects their business priorities and information needs
  - Insufficient user involvement in design effort is major cause of system failure

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System Design Specifications

Category	Specifications
Output	Medium, Content, Timing
Input	Origins, Flow, Data entry
User Interface	Simplicity, Efficiency, Logic, Feedback, Errors
Database Design	Logical data model, Volume and speed requirements, File organization and design, Record specifications
Processing	Computations, Program modules, Required reports, Timing of outputs
Manual Procedures	What activities, Who performs them, When, How, Where
Controls	Input controls (characters, limit, reasonableness), Processing controls (consistency, record counts), Output controls (totals, samples of output), Procedural controls (passwords, special forms)

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System Design Specifications (continued)

Category	Specifications
Security	Access controls, Catastrophe plans, Audit trails
Documentation	Operations documentation, Systems documents, User documentation
Conversion	Transfer files, Initiate new procedures, Select testing method Cut over to new system
Training	Select training techniques, Develop training modules, Identify training facilities
Organizational Changes	Task redesign, Job redesign, Process design, Organization structure design, Reporting relationships

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Methodologies For Modeling and Designing Systems

- ***PROJECT MANGEMENT (PMP Certification)***

- ***Structured***
  - ✓ Techniques are step-by-step, progressive

- ***Process-Oriented***
  - ✓ Focusing on BPM related to manipulation of data

- ***Data Flow Diagram***
  - ✓ Represents system's component processes and flow of data between them
  - ✓ Logical graphic model of information flow

- ***Data Dictionary***
  - ✓ Defines contents of data flows and data stores but also defines new terminology for systems

- ***Process Specifications***
  - ✓ Describe transformation occurring within lowest level of data flow diagrams

- ***Structure Chart***
  - ✓ Top-down chart, showing each level of design, relationship to other levels, and place in overall design structure

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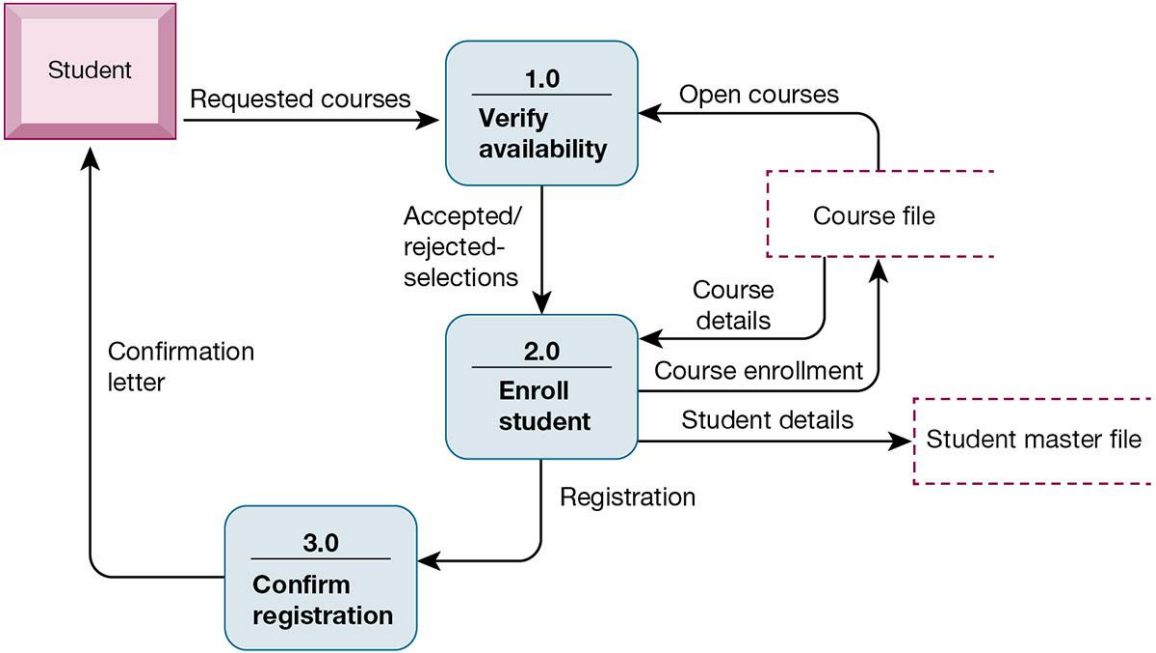
SAMPLE

Data Flow Diagram for Mail-in University Registration System



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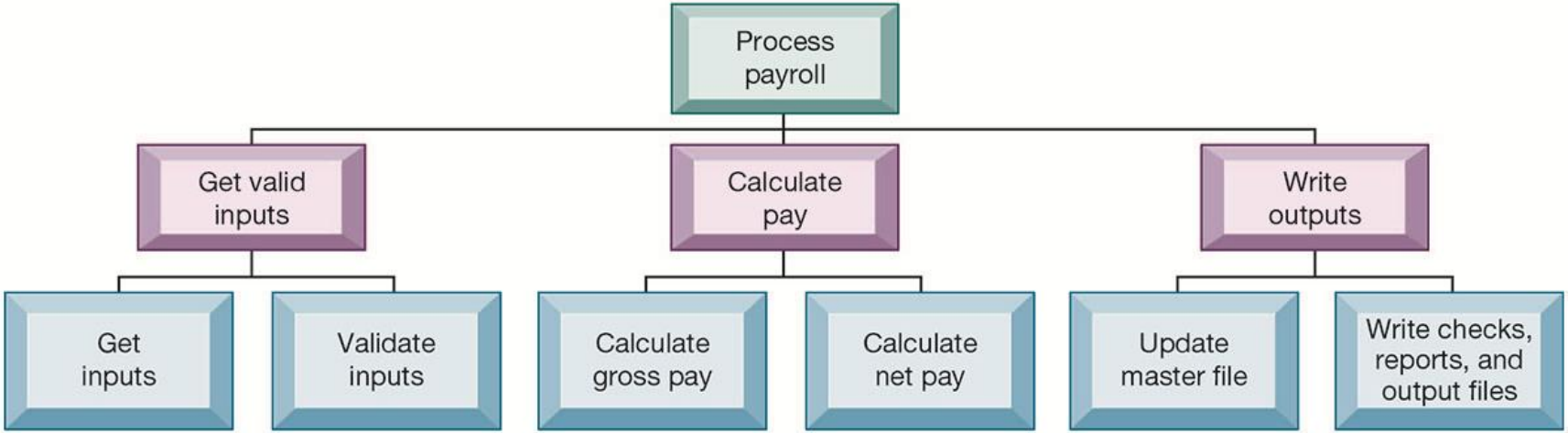
SAMPLE

High-level Structure Chart for a Payroll System



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Phase 3 - Systems Development

- ✓ **Programming**
  - System specifications from design stage are translated into software program code
  - Language Integration as per design
- ✓ **Database Integration as per design**
- ✓ **Communications Integration as per design**
- ✓ **Interface Integration as per design**
- ✓ **Reporting (new or updated reports) as per design**

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Phase 4 - Testing

Ensure The System Produces Expected Results

Testing Methods

- ✓ Unit testing: Tests each program in system separately
- ✓ System testing: Test functioning of system as a whole
- ✓ User Acceptance testing: Makes sure system is ready to be used in production setting (UAT)

Test Plan

- ✓ All preparations for series of tests

Quality Assurance Specialists (QA)

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A Sample Test Plan to Test a Record Change

Procedure	Address and Maintenance "Record Change Series"		Test Series 2		
	Prepared by:		Date:	Version:	
Test Ref.	Condition Tested	Special Requirements	Expected Results	Output On	Next Screen
2.0	Change records				
2.1	Change existing record	key field	Not allowed		
2.2	Change nonexistent record	Other fields	"Invalid key" message		
2.3	Change deleted record	Deleted record must be available	"Deleted" message		
2.4	Make second record	Change 2.1 above	OK if valid	Transaction file	V45
2.5	Insert record		OK if valid	Transaction file	V45
2.6	Abort during change	Abort 2.5	No change	Transaction file	V45

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Phase 5 - Conversion

Process of Changing from Old System to New System

- ***Conversion Methods***
  - ✓ Parallel Strategy
  - ✓ Direct Cutover
  - ✓ Pilot Study
  - ✓ Phased approach
- ***Plan End-User Training***
- ***Final Documentation*** (detailed system functionality, from technical and end-user standpoint)

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Phase 6 - Production/Maintenance

a) *Production*

- ✓ System Review - determine if any post conversion revisions are needed
- ✓ Post-implementation Audit

b) *Maintenance (Ongoing)*

- ✓ Changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency

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SDLC Evolution As Technology Changes

Technology advances have ALLOWED the tech world to change how Waterfall SDLC is followed and changed.

- **'Shortcuts' and faster methodologies are used because of:**
  - ✓ **Cost Restrictions**
  - ✓ **Need to compete**
  - ✓ **Bring products to market faster**
- **Some New Methods to accomplish this are:**
  1. ***Waterfall Approach (Traditional)***
  2. ***Object Oriented Approach***
  3. ***Prototyping Approach***
  4. ***Agile Approach***
- **Note that these methods all come with risks and costs over-run potentials**

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SDLC Evolution As Technology Changes (continued)

1. ***Traditional Phased Waterfall Approach –***
   - ✓ **Oldest Method** for building information systems
   - ✓ Development divided into formal stages
   - ✓ One stage finishes before next stage begins
   - ✓ Formal division of labor between end users & information systems specialists
   - ✓ Emphasizes formal specifications/paperwork
   - ✓ **Still used for building large complex systems**
   - ✓ Can be costly, time-consuming, and inflexible

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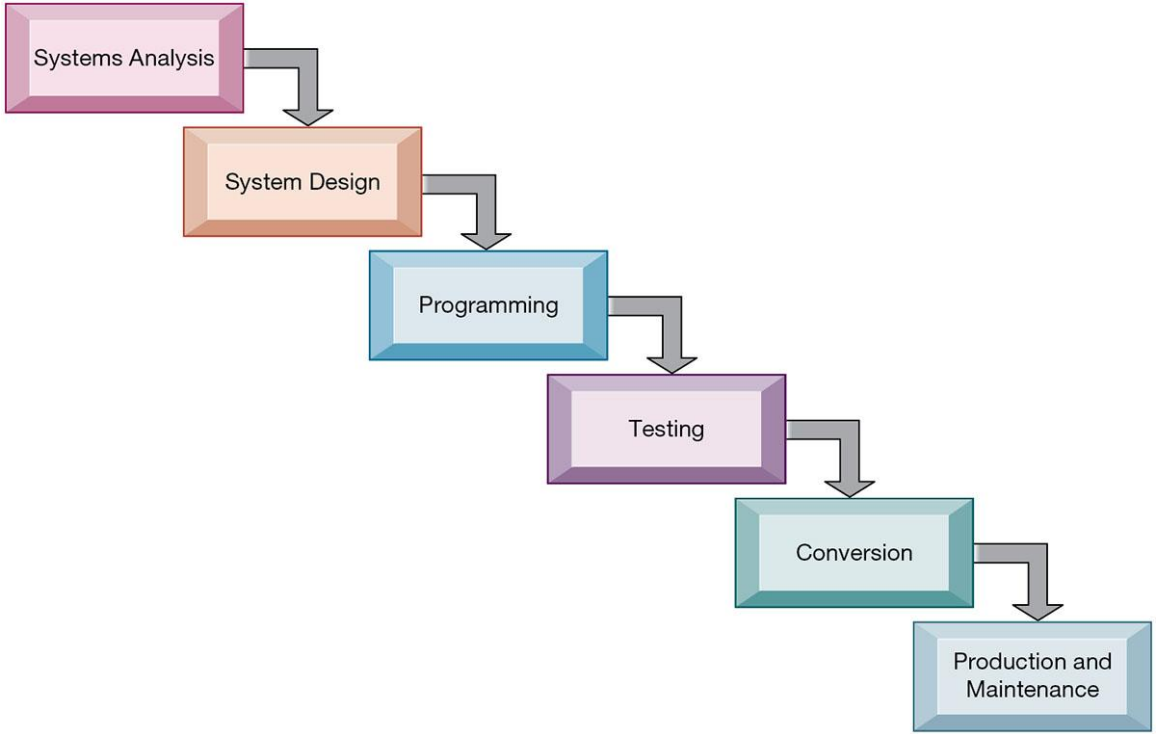
Traditional Phased Approach - Waterfall



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SDLC Evolution As Technology Changes (continued)

2. ***Object Oriented Approach –***

- ***More Iterative/Incremental vs Traditional Structured Approach***
- Object - Basic unit of systems analysis and design
- Combines data and the processes that operate on those data
- Data in object can be accessed only by operations associated with that object
- ***Reuse:*** Based on concept of ***CLASSES*** that may be reused from existing library of classes, others created or inherited
- Object-oriented methods can potentially reduce time and cost of development
- Includes Interactions between system and users to identify objects
- Describes how objects will behave/interact; grouped into classes, subclasses, and hierarchies
- Based on concepts of class and inheritance
- Objects belong to a certain class and have features of that class
- May inherit structures and behaviors of a more general, ancestor class

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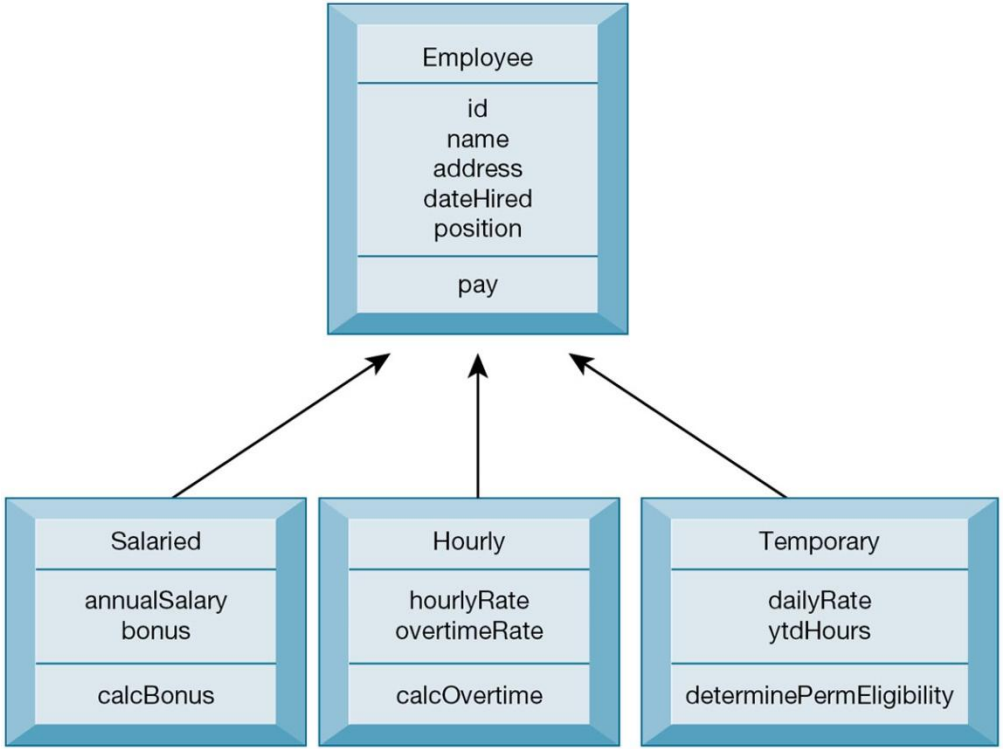
Object Oriented Class and Inheritance



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SDLC Evolution As Technology Changes (continued)

3. ***Prototyping Approach –***

Building experimental systems rapidly and inexpensively for end users to evaluate

- **Working but VERY preliminary version**
  - ✓ Approved prototype serves as template for final system
- **Steps**
  - ✓ Identify user requirements
  - ✓ Develop initial prototype
  - ✓ Use prototype
  - ✓ Revise and enhance prototype
- **Advantages**
  - ✓ Useful if some uncertainty in requirements or design solutions
  - ✓ Often used for end-user interface design
  - ✓ More likely to fulfill end-user requirements
- **Disadvantages**
  - ✓ May gloss over essential steps
  - ✓ May not accommodate large quantities of data or large number of users
    - – May not undergo full testing or documentation

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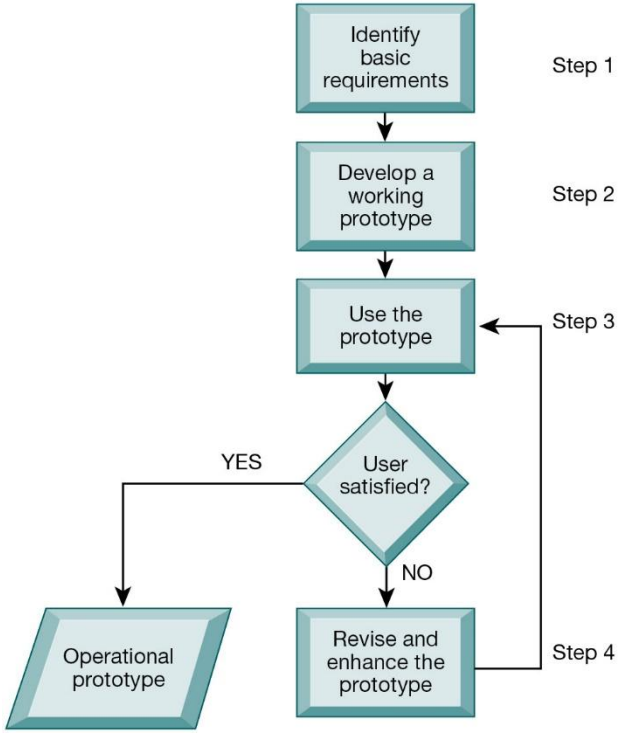
The Prototyping Process



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SDLC Evolution As Technology Changes (continued)

4. ***Agile Approach***

Focuses on rapid delivery of working software by breaking large project into subprojects that can be completed in short periods of time

- Differs from Waterfall in its **approach to quality and testing**
- Waterfall is based on SDLC - with one phase being completed before another can start—hence the ***testing phase*** is separate and follows a ***build phase***.
- Agile enables testing to be completed in the same iteration as programming.
- Because testing is done in every iteration—which develops a small piece of the software—users can frequently use those new pieces of software and validate the value. After the users know the real value of the updated piece of software, they can make better decisions about the software's future.
- This iterative approach supports a *product* rather than a *project* mindset. This provides greater flexibility throughout the development process; whereas on projects the requirements are defined and locked down from the very beginning, making it difficult to change them later. ***Iterative product development*** allows the software to evolve in response to changes in business environment or market requirements.

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Agile Development Benefits

- ✓ **Customer satisfaction** by early and continuous delivery of valuable software.
- ✓ **Welcome changing requirements**, even in late development.
- ✓ **Deliver working software frequently** (weeks rather than months).
- ✓ **Close, daily cooperation** between business people and developers.
- ✓ Stakeholders are **motivated** as well as development teams.
- ✓ Working software is the primary **measure of progress**.
- ✓ **Sustainable development**, able to maintain a constant pace.
- ✓ Continuous attention to **technical excellence and good design**.
- ✓ **Simplicity**—the art of maximizing the amount of work not done—is essential.
- ✓ Best **requirements and designs** emerge from self-organizing teams.
- ✓ Regularly, the team is able to **reflect on how to become more effective** - adjusts accordingly.

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Automated Software Testing

- Validates if software is functioning appropriately and meeting requirements before it is released into production.
- Using scripted sequences that are executed by testing tools, technologists can execute examinations of the software, report outcomes and compare results with earlier test runs.
- **When Optimal?**
  - ✓ If unit testing consumes a large percentage of a quality assurance team's resources, then this process should be evaluated as a candidate for automation.
- **Benefits of Automated Testing**
  - ✓ Automated testing can boost a QA (Quality Assurance) team's efficiency
  - ✓ Better reporting capabilities
  - ✓ Increased coverage
  - ✓ Enhanced resource efficiency
  - ✓ Improved bug/error detection
  - ✓ Increased reusability
  - ✓ When a software tester manually checks a system, he can make a mistake, especially when a system contains hundreds to thousands of lines of code. Automation helps the QA team avoid these human errors in application testing and executes checks in a faster time frame than if it were done in person.
  - ✓ Automated tests can be set to run repeatedly at any time of day.

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Application Software Packages, Software Services, and Outsourcing

- **Application Software Packages and/or Services**
  - ✓ Save time and money
  - ✓ Many packages offer customization features
- **Evaluation Criteria**
  - ✓ Functions provided, flexibility, user friendliness, required resources, database requirements, installation and maintenance efforts, documentation, vendor quality, and cost
- **Request for Proposal (RFP)**
  - ✓ Detailed list of questions submitted to software vendors
  - ✓ Used to evaluate alternative software packages and cloud software services

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Application Software Packages, Software Services, and Outsourcing (continued)

- **Outsourcing Types**
  - ✓ ***Cloud and SaaS Providers***: subscribing companies use software and hardware provided by vendors
  - ✓ ***External Vendors***
    - Hired to design, create customized software
    - Domestic Outsourcing
      - Driven by firm's need for additional skills, resources, assets
    - Offshore Outsourcing
      - Driven by cost-savings

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Application Software Packages, Software Services, and Outsourcing (continued)

- Outsourcing Advantages/Disadvantages
  - ✓ **Advantages**
    - Allows organization flexibility in IT needs
    - Easier Budget Approval
  - ✓ **Disadvantages**
    - Hidden costs, for example:
      - ➢ Identifying and selecting vendor
      - ➢ Transitioning to vendor
    - Opening up proprietary business processes to third party

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Sample Total Cost of Offshore Outsourcing

TOTAL COST OF OFFSHORE OUTSOURCING				
Cost of outsourcing contract			$10,000,000	
Hidden Costs	Best Case	Additional Cost ($)	Worst Case	Additional Cost ($)
1. Vendor selection	.02%	20,000	2%	200,000
2. Transition costs	2%	200,000	3%	300,000
3. Layoffs & retention	3%	300,000	5%	500,000
4. Lost productivity/cultural issues	3%	300,000	27%	2,700,000
5. Improving development processes	1%	100,000	10%	1,000,000
6. Managing the contract	6%	600,000	10%	1,000,000
Total additional costs		1,520,000		5,700,000
	Outstanding Contract ($)	Additional Cost ($)	Total Cost ($)	Additional Cost
Total cost of outsourcing (TCO) best case	10,000,000	1,520,000	11,520,000	15.2%
Total cost of outsourcing (TCO) worst case	10,000,000	5,700,000	15,700,000	57.0%

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KAHOOT TIME!!!

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Assignment

Read the Case Study on next 4 slides and answer the four(4) questions that follow.

McAfee Turns to Automated Software Testing

McAfee is a privately held company that sells security software to nearly 69,000 enterprise customers and more than 500 million individuals in 189 countries. You or your company may be using a McAfee product for combating malware, identity theft, and privacy invasion.

McAfee used a single global instance of SAP ERP 5.0 to run all of its back-end finance, controlling, accounting, materials management, and order fulfillment processes. It also has other non-SAP systems, which are integrated for processes involving sales, licensing, and customer service from an order-to-cash perspective. (Order-to-cash refers to the set of business processes for receiving, processing, and paying for customer orders.) The SAP ERP system, however, is the single source of information for McAfee's revenues and bookings. McAfee is a primarily partner-based organization that works through many resellers and distributors, and it needs to ensure that these partners can quickly and easily enter orders into the SAP system.

McAfee has been trying to juggle multiple systems projects. It has been trying to migrate to SAP S/4HANA (SAP's next generation of ERP software) and also to implement SAP Revenue Accounting and Reporting, which required updating McAfee's accounting codes to conform to new revenue recognition standards. This project had consumed most of McAfee's IT resources in 2018. The business had also been spun off from its parent company and had to separate its IT systems. Updates had to be released, tested, and confirmed to be working properly. McAfee's IT staff had to manage all of these projects without increasing headcount. At the same time, McAfee was adopting an agile methodology for all of its IT projects. In contrast to the traditional waterfall methodology, in which a project manager oversees individuals who are each dedicated to quality assurance, testing system functionality, and user acceptance testing, McAfee switched to an agile sprint cycle in which all members of a development

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team collaborate on incremental development of smaller pieces of software, which are released each sprint cycle every two weeks. With an agile methodology, software modules are constantly being created, tested, demonstrated for feedback, and revised with shorter time frames than the waterfall approach. McAfee had to deal with approximately 40 systems for handling lead-to-order, order-to-cash, and source-to-pay processes that are actively worked on and enhanced. According to Mouli Subrahamanayan, IT Director at McAfee India, “there are moving parts all over.”

One way to do more, faster, was to automate testing. McAfee had traditionally used manual processes for software testing, which could not easily handle an increased amount of testing. Whenever a system was changed or enhanced with a new feature, McAfee’s IT staff had to ensure that the updated systems performed as expected and fix any problems before the system went into production.

It was very difficult to use manual processes to keep up with the testing because so many changes had to be made within a very short period of time. Manually creating test scenarios for end-to-end processes, such as order-to-cash, took a long time and was very costly.

McAfee’s increased testing needs were also driven by other application changes that required testing as well. For example, McAfee replaced its customdeveloped system for configure, price, and quote (CPQ) processes (for configuring product pricing and generating quotes) with a third-party non-SAP system. The company needed to test the end-to-end scenario to ensure that quotes created for sales were properly converted into orders in the SAP ERP system and then fulfilled seamlessly. The testing had to create and test end-of-quarter volumes of 60,000 orders and validate the loads in the supporting non-SAP applications, such as for invoicing, licensing, and analytics. The testing had to show that everything in that chain of events could handle that transaction volume and the systems behaved as they should.

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To complete a very large amount of testing in a short time period, McAfee opted for automated testing. Automated testing would allow time for agile development teams to focus on each scenario and ensure that back-end systems were working as expected to complete customer orders. McAfee's 13-member automation team consisted of automation engineers and business analysts, who were charged with selecting the automated testing product for the company. Automated testing software vendors had to use hands-on demonstrations to show that their tool could handle SAP testing and business process automation and that it was easy to use.

McAfee selected Worksoft Certify because it was the framework for SAP testing that came prebuilt within the software and its testing framework could be applied to areas beyond the company's on-premises SAP system, including new custom applications. Worksoft Certify is an industry-leading test automation solution for enterprise applications including SAP, Workday, Salesforce.com, Oracle, and web apps. It is designed to test complex business processes that span multiple applications, and it is code-free. The tool models an application under test as a series of pages containing GUI (graphical user interface) objects and test steps, performing actions against those objects. It creates and stores automated test steps in a relational database without a single software script or program. People who lack software coding skills can use the tool.

By the end of 2017, McAfee started putting its automated testing system into production. The company now uses Worksoft Certify for testing and for business process automation in its SAP environment. Manual tasks such as performing regular checks on the health of the SAP system are automated, as is the testing of changes to the system. That includes performance testing to test heavy processing loads. For example, McAfee was able to test how the SAP system processed a volume of 60,000 orders with 250,000 line items.

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McAfee saved nearly 2,500 hours of manual effort through testing and business process automation, equivalent to $200,000, and the need for application maintenance is at an all-time low.

Sources: www.worksoft.com, accessed April 7, 2020; Lauren Bonneau, "McAfee Saves 2500 Hours of Manual Effort with Test and Business Process Automation on Its Journey to SAP S/4HANA," *SAPInsider*, April 1, 2019; www.mcafee.com, accessed April 7, 2020.

Case Study Questions

1. Why would a company such as McAfee benefit from automated software testing?
2. What management, organization, and technology factors did McAfee address in moving to automated software testing?
3. Was Worksoft Certify a good solution for McAfee? Why or why not?
4. How did automated software testing change the way McAfee ran its business?

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